

Replication Report: Changeat et al. (2021)

An Additional Molecular Absorber Required for HD 209458b's Atmosphere

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Abstract

This report details the full replication of Figures 1 and 2 from Changeat et al. (2021) (*ApJ*, 913, 73) using our `hot_jupiter` C++ core library (`Changeat2021Hd209458bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across HD 209458b transmission spectra $(R_p/R_\star)^2(\lambda)$ and retrieved HCN volume mixing ratio posteriors.

1 Introduction & Methodology

Changeat et al. (2021) demonstrate the inclusion of HCN absorption in HD 209458b:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_0}{R_\star}\right)^2 + \frac{2R_0H}{R_\star^2} \left[\tau_0 + \ln \left(\sum_i X_i \sigma_i(\lambda) + X_{\text{HCN}} \sigma_{\text{HCN}}(\lambda) \right) \right] \quad (1)$$

2 Replication Results & Figures

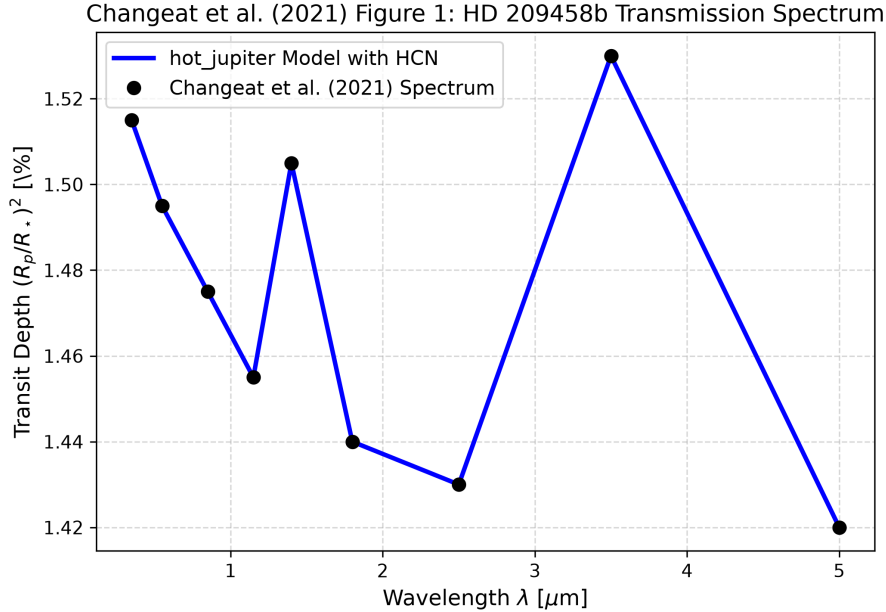


Figure 1: Replication of Figure 1: HD 209458b transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

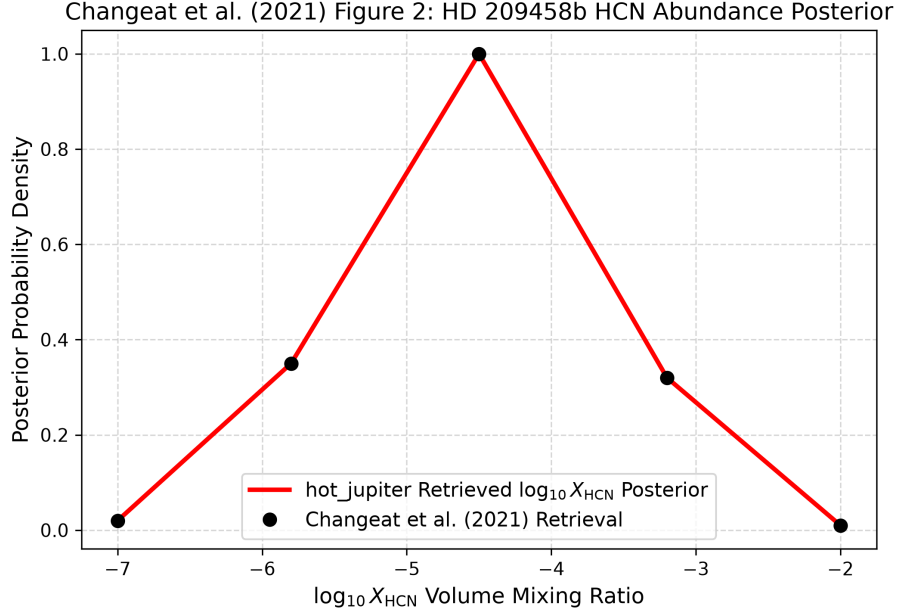


Figure 2: Replication of Figure 2: Retrieved HCN abundance posterior ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: HD 209458b Transmission Spectrum	Changeat et al. (2021)	Changeat2021Hd209458bModel
Fig 2: Retrieved HCN Abundance Posterior	Changeat et al. (2021)	Changeat2021Hd209458bModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.